

PCA- and SVM-Grad-CAM for Convolutional Neural Networks: Closed-form Jacobian Expression

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Abstract—Convolutional Neural Networks (CNNs) are an effective approach for classification tasks, particularly when the training dataset is large. Although CNNs have long been considered a black-box classification method, they can be used as a white-box method through visualization techniques such as Grad-CAM. When training samples are limited, incorporating a Principal Component Analysis (PCA) layer and/or a Support Vector Machine (SVM) classifier into a CNN can effectively improve classification performance. However, traditional Grad-CAM cannot be directly applied to PCA and/or SVM layers. It is important to generate attention regions for PCA and/or SVM layers in CNNs to facilitate the development of white-box methods. Therefore, we propose “PCA-Grad-CAM”, a method for visualizing attention regions in PCA feature vectors, and “SVM-Grad-CAM”, a method for visualizing attention regions in an SVM classifier layer. To complete our methods analytically, it is necessary to solve the closed-form Jacobian consisting of partial derivatives from the last convolutional layer to the PCA and/or SVM layers. In this paper, we present the exact closed-form Jacobian and the visualization results of our methods applied to several major datasets.

Index Terms—Grad-CAM, Convolutional Neural Network, Principal Components Analysis, Support Vector Machine, Jacobian

I. INTRODUCTION

Since Convolutional Neural Networks (CNNs) are an effective approach for various classification tasks, they have been widely used in diverse fields, such as medical diagnosis [1], [2], agriculture [3], [4], industrial applications [5], [6], and others. At the time AlexNet [7], which marked the beginning of CNN’s widespread adoption, was introduced in 2012, there was no way to determine which attention regions of the image contributed to the model’s predictions. Grad-CAM, proposed by Selvaraju et al. [8] in 2017, enabled the visualization of attention regions in CNNs. That method is still widely used in various studies, including [9], [10], and [11]. When using CNNs in medical applications, it is recommended to use visualization methods such as Grad-CAM [12].

Even though Grad-CAM can only be applied to plain CNNs, modern architectures often include specialized components, such as principal component analysis (PCA) and/or support vector machine (SVM) layers, to enhance performance. There is some evidence that these ingenuities lead to improved performance. For example, Jean et al. [13] reported an improvement in accuracy from 95.64% to 98.32% by implementing an SVM layer into a CNN for classifying cocoa beans. Similar findings have also been reported for tasks such as the detection of cervical cancer cells [14] and microvascular morphological type recognition [15]. Moreover, several studies have implemented both PCA and SVM into CNNs. Mahmood

et al. [16] reported that applying both PCA and SVM to CNNs improved accuracy from 92.6% to 94.7% for the Caltech-101 classification task. Similar findings have also been reported for tasks such as facial recognition [17], hand gesture recognition [18], face verification [19], oil rig classification [20], and epilepsy detection [21]. As we can see from these studies, applying both PCA and SVM to CNNs is a popular approach. However, the conventional Grad-CAM [8] cannot be applied to these models; that is, they function as black-box models.

Therefore, we propose “PCA-Grad-CAM” and “SVM-Grad-CAM” to visualize the focus regions of PCA and SVM layers embedded in CNNs, respectively. An overview comparing conventional Grad-CAM, PCA-Grad-CAM, and SVM-Grad-CAM is shown in Figure 1. “Conventional-Grad-CAM” proposed by Selvaraju et al. [8] can only be applied to plain CNNs. Unlike this, PCA-Grad-CAM is a method for visualizing the focus regions of each principal component; that is, if the number of principal components is B , then the number of PCA-Grad-CAM visualizations is also B . As shown in Figure 1, P_1^ω , where $\omega \in \{\oplus, \ominus\}$, represents the PCA-Grad-CAM corresponding to the first dimension of the PCA feature vector. $P_1^\oplus$ represents the region that positively influences the first principal component, while $P_1^\ominus$ represents the region that negatively influences it. Similarly, P_2^ω where $\omega \in \{\oplus, \ominus\}$ corresponds to the second principal component of the PCA feature vector. From $P_1^\oplus$ and $P_2^\ominus$ in Figure 1, it can be inferred that the cat’s mouth positively influences the first principal component, while the cat’s ear negatively influences the second principal component.

Since there are no existing methods for visualizing the focus regions of each principal component in CNNs, the proposed method, PCA-Grad-CAM, could be considered a potentially useful approach. In contrast, the second proposed method, SVM-Grad-CAM, is designed to visualize the focus region of an SVM layer embedded in CNNs. As described in the previous paragraph, combining a CNN with an SVM is an influential approach; however, such a model functions as a black box. Since SVM-Grad-CAM aims to address this issue, it may serve as a helpful tool for improving model interpretability.

Additionally, gradients of certain functions in CNNs, PCA, and SVM are required for the mathematical representation of PCA- and SVM-Grad-CAMs. Since these are multi-input multi-output functions, their gradients can be represented in Jacobian form. This paper provides exact closed-form expressions for these Jacobians. Deriving a closed-form solution means expressing the result explicitly, as if exposing every component of the equation to full view, without relying on

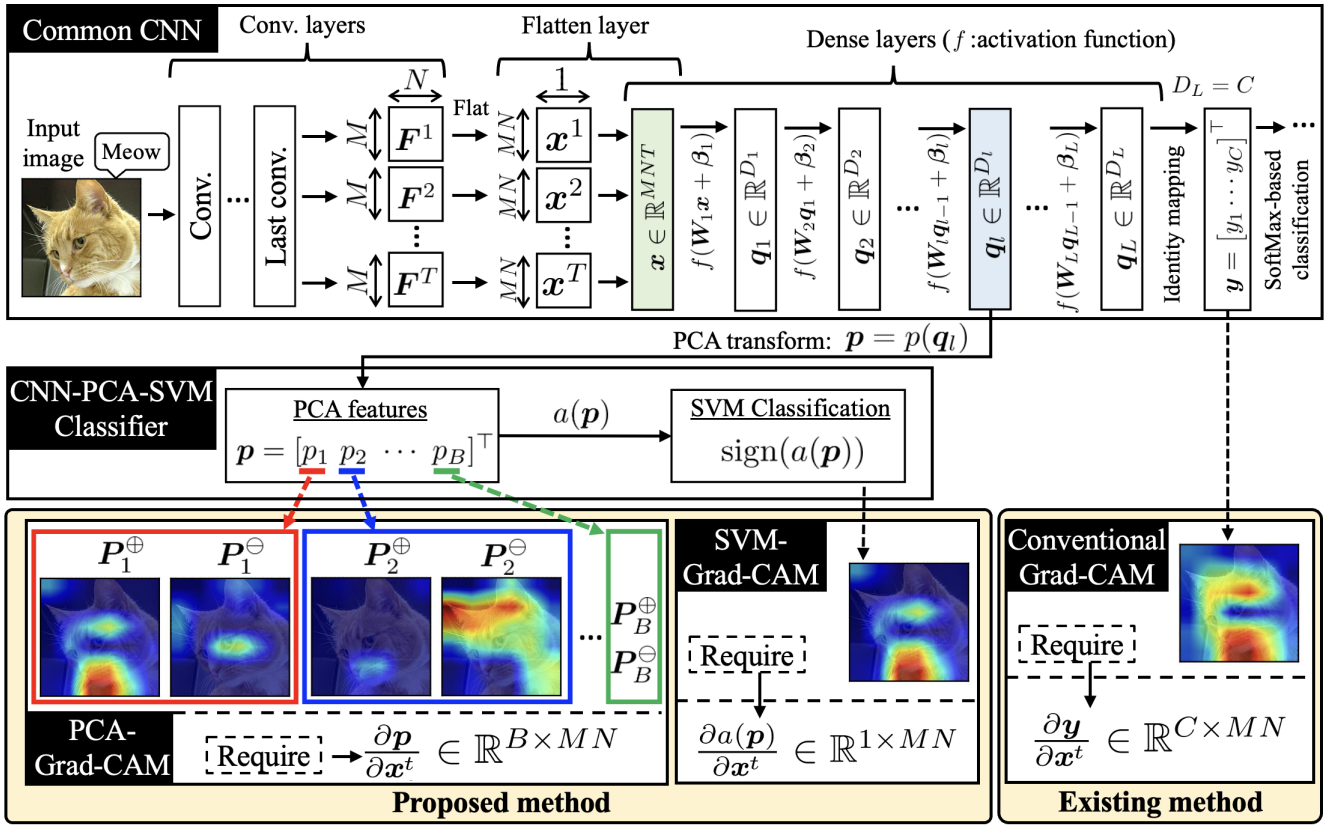


Fig. 1. Overview of the existing method (conventional-Grad-CAM [8]) and the proposed methods (PCA- and SVM-Grad-CAM).

hidden computations or iterative approximations. Therefore, we also provide some theoretical analysis based on these closed-form expressions.

II. RELATED WORKS

CAM, proposed by Zhou et al. [22], was a pioneering visualization method for CNNs and has been applied in many studies, e.g., [23], [24], [25]. However, this method can only be applied to a limited set of CNNs that include a global average pooling layer and a classifier layer immediately after the last convolutional layer. To overcome this limitation, Grad-CAM was proposed by Selvaraju et al. [8]. Since Grad-CAM can be applied to various CNN architectures, it has become a widely used method. Moreover, in addition to Grad-CAM, various CAM-based methods have been proposed, as shown in Table I. In this section, we first introduce these methods, and then describe the novelty and academic contributions of our proposed approach.

A. Gradient-based CAM

Methods for visualizing the focus regions of CNNs can be divided into gradient-based and gradient-free approaches. We first introduce gradient-based methods, including Grad-CAM++ [26], XGrad-CAM [30], Relevance-CAM [34], HiRes-CAM [38], and Layer-CAM [42].

Grad-CAM++ [26] is a method that applies higher-order derivatives to compute the weights of feature maps. Grad-CAM++ is superior to Grad-CAM with respect to object

detection of a target class. XGrad-CAM [30] is a method that uses both activation values and gradients of feature maps to compute the feature map weights. Grad-CAM uses the sum of the gradients of a feature map as its weight, whereas XGrad-CAM computes the element-wise product of the feature map and its gradients, and then takes the sum as the weight. Relevance-CAM [34] uses the sum of the relevance map obtained through relevance propagation as the weight for each feature map. It can visualize not only the last convolutional layer but also all convolutional layers, including the shallow ones. HiRes-CAM [38] is a CAM method that uses the Hadamard product of the feature map and its gradients. HiRes-CAM is superior to Grad-CAM in terms of the level of detail in the focus regions. Layer-CAM [42], which uses a weighting similar to HiRes-CAM, aims to visualize shallower layers. As shown in the ‘‘Applied cases’’ in Table I, the gradient-based CAMs introduced here have been applied to various application studies.

Note that in gradient-based methods, the generated CAM can sometimes change significantly with a slight change in the input. One method to address this issue is to average the CAMs generated from multiple outputs obtained by randomly noised inputs. This approach is called SmoothGrad [63], and it leads to more stable results. A representative method that incorporates this approach is Smooth Grad-CAM++ [64].

TABLE I
CAM FAMILY COMPRISING VARIOUS APPROACHES.

Method [ref. no.]	Year	Citation*	Gradient?	Applied cases (detection or prediction target)
CAM [22] (Pioneer)	2015	13,409	No	Food localization [23], Diabetic retinopathy [24], Building defects [25]
Grad-CAM [8]	2016	31,032	Yes	Fall armyworm [9], Microscopic algae [10], Gas-liquid jet [11]
Grad-CAM++ [26]	2018	3,748	Yes	Lung Cancer [27], Brain tumor [28], Malignant skin diseases [29]
XGrad-CAM [30]	2020	397	Yes	SARS-COVID [31], Brain tumor [32], Specific emitter identification [33]
Relevance-CAM [34]	2021	101	Yes	Vertebral compression fracture [35], Small bowel [36], Cities identification [37]
HiRes-CAM [38]	2020	174	Yes	Endoscopic image [39], Insect pests [40], Ship detection [41]
Layer-CAM [42]	2021	811	Yes	Aerial scene [43], Pneumoconiosis detection [44], Satellite images [45]
Score-CAM [46]	2020	1,410	No	Plant disease [47], [48], Skin lesion [49], Genotype [50]
Ablation-CAM [51]	2020	590	No	Plant disease [52], Graphene strength [53], Pulmonary edema [54], Medical image [55]
Shap-CAM [56]	2022	54	No	-
Shaplay-CAM [57]	2025	3	No	-
LIFT-CAM [58]	2021	125	No	Skin diseases [59]
Eigen-CAM [60]	2020	545	No	Medical image [55], Periodontal bone loss severity [61], Power grid fault location [62]

*: Total citations of the papers proposing each method. These values were verified by Google Scholar on 18 July 2025.

B. Gradient-free CAM

Next, we introduce gradient-free CAMs, such as Score-CAM [46], Ablation-CAM [51], Shap-CAM [56], Shapley-CAM [57], LIFT-CAM [58], and Eigen-CAM [60]. Among these, Score-CAM [46] is the most representative method. The key step in generating Score-CAM is creating a mask image by overlaying a feature map onto the input image. By inputting this mask image into the trained CNN, the network outputs a class score, which is then used as the weight for the corresponding feature map. As shown in Table I, Score-CAM is the most commonly used method among gradient-free CAMs. Another approach is Ablation-CAM [51]. The key step is to compute the change in the estimated score by replacing feature map values with zero or their average. The size of the change in score is used as the weight for the corresponding feature map. Ablation-CAM is a simple and fast method.

Shap-CAM [56] and Shapley-CAM [57] are methods that use the contribution of each pixel to the class prediction, based on Shapley values [65], to determine the weight of each feature map. Although these methods require a high computational cost, they can estimate contributions more accurately than gradient-based methods. LIFT-CAM [58] is a method that employs an approximate approach to compute Shapley values. These methods are technically interesting because they introduce game theory into CAM.

Additionally, there are CAM methods that summarize all feature maps. For example, Eigen-CAM [60] is generated by applying PCA to the vectorized feature maps. By generating only the first principal component, we can identify the essential focus region. Note that Eigen-CAM [60] does not reflect the contribution to class prediction. Although both Eigen-CAM and our proposed PCA-Grad-CAM use PCA, they are fundamentally different methods. The main purpose of Eigen-CAM is to extract the principal components from all feature maps. In contrast, the main purpose of PCA-Grad-CAM is to visualize the focus regions corresponding to each principal component in any dense layer. That is, PCA-Grad-CAM generates Grad-CAMs for each dimension of the PCA-transformed feature vector in any dense layer.

The gradient-free CAMs introduced in this subsection have been applied in various studies, as shown in the “Applied cases” in Table I. As an exception, application cases of Shapley

value-based CAMs are rare.

C. Positioning of this study

The citation counts of the CAM methods shown in Table I indicate that gradient-based CAMs are more popular than gradient-free CAMs. This tendency is clearly observed in game theory-based CAMs (Shap-CAM [56], Shapley-CAM [57], and LIFT-CAM [58]). Although these methods are interesting, due to factors such as implementation difficulty and high computational cost, their applied cases may be limited. In contrast, the most popular method is Grad-CAM, with total citation counts exceeding 30,000. Therefore, we focus on extending Grad-CAM.

As previously mentioned in Section I, Grad-CAM is a superior method for visualizing CNNs; however, it cannot be applied to PCA and SVM layers integrated into CNNs. Moreover, the various CAMs shown in Table I also cannot be applied to PCA and SVM layers. Additionally, as previously mentioned in Subsection II-B, Eigen-CAM is also not applicable to these layers. Therefore, by extending the theory proposed by Selvaraju et al. [8], we propose a novel Grad-CAM technique applicable to PCA and SVM layers integrated into CNNs. By using our methods, PCA-Grad-CAM and SVM-Grad-CAM, the focus regions corresponding to each principal component and the SVM decision can be clearly visualized. Therefore, we believe our study provides an academic contribution.

III. PRELIMINARY KNOWLEDGE

This section introduces the background knowledge needed to understand the proposed method. Basic mathematical notation is given in Appendix A.

A. Conventional CNN

The CNN model architecture assumed in our study is shown at the top of Figure 1. An input image is processed through several convolutional layers, resulting in T feature maps $\mathbf{F}^1, \dots, \mathbf{F}^T$. Herein, let us denote the t -th feature map by $\mathbf{F}^t = [\mathbf{f}_{m,n}^t] \in \mathbb{R}^{M \times N}$, $t \in \mathbb{N}_{\leq T}$, which is represented in matrix form. Where M , N , and T represent the vertical size, horizontal size, and total number of feature maps, respectively.

By a flatten layer in CNNs, the matrix-form feature map $\mathbf{F}^t$ is transformed into a vector-form

$$\mathbf{x}^t = [f_{1,1}^t \ f_{1,2}^t \ \cdots \ f_{M,N}^t]^\top = [x_1^t \ x_2^t \ \cdots \ x_{MN}^t]^\top \in \mathbb{R}^{MN}. \quad (1)$$

By applying this transformation to $\forall t \in \mathbb{N}_{\leq T}$, we concatenate the resulting vectors into

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}^1 \\ \vdots \\ \mathbf{x}^T \end{bmatrix} \in \mathbb{R}^{MNT}. \quad (2)$$

This is generally referred to as the CNN feature vector. As show at the top of Figure 1, the CNN feature vector $\mathbf{x}$ is used for classification through L fully connected (dense) layers, as follows:

$$\begin{aligned} \mathbf{q}_1 &= f(\mathbf{W}_1 \mathbf{x} + \beta_1) \in \mathbb{R}^{D_1}, \\ \mathbf{q}_2 &= f(\mathbf{W}_2 \mathbf{q}_1 + \beta_2) \in \mathbb{R}^{D_2}, \\ &\vdots \\ \mathbf{q}_l &= f(\mathbf{W}_l \mathbf{q}_{l-1} + \beta_l) \in \mathbb{R}^{D_l}, \\ &\vdots \\ \mathbf{q}_{L-1} &= f(\mathbf{W}_{L-1} \mathbf{q}_{L-2} + \beta_{L-1}) \in \mathbb{R}^{D_{L-1}}, \\ \mathbf{q}_L &= f(\mathbf{W}_L \mathbf{q}_{L-1} + \beta_L) \in \mathbb{R}^{D_L}, \\ \mathbf{y} &= \mathbf{q}_L \in \mathbb{R}^C, \quad C = D_L, \end{aligned} \quad (3)$$

where, $\mathbf{W}_l \in \mathbb{R}^{D_l \times D_{l-1}}$ and $\beta_l \in \mathbb{R}^{D_l}$, $D_0 = MNT$. $\mathbf{W}_l$ and β_l are the parameters for the affine transformation of the l -th dense layer, and f represents the activation function. The output vector is $\mathbf{y} = [y_1 \cdots y_C]^\top$, and the predicted class is given by $\arg\max_c y_c$.

B. PCA layer

In our study, we assume that PCA is applied to the l -th dense layer, $\mathbf{q}_l = f(\mathbf{W}_l \mathbf{q}_{l-1} + \beta_l)$. As shown in ‘‘PCA transform’’ in Figure 1, when we denote PCA function by $p(\cdot)$, the PCA feature vector can be represented as $\mathbf{p} = p(\mathbf{q}_l)$. In general, PCA is used for dimensionality reduction. Therefore, when we denote the dimension of the PCA feature vector by B , the relationship $B \leq D_l$ holds. That is, the PCA feature vector can be represented as $\mathbf{p} = [p_1 \ p_2 \ \cdots \ p_B]^\top$.

Next, we explain the details of the PCA function $p(\cdot)$. To obtain this function, the covariance matrix composed of observation samples is required. Herein, we assume that there are R input images, and denote the vector $\mathbf{q}_l$ generated from the r -th input image by $\mathbf{q}_l^r$. In this case, the observation data matrix $\mathbf{Q}$, its zero-mean version $\mathbf{Q} - \mathbf{U}$, the unbiased covariance matrix Σ , and the eigenvalue diagonal matrix Λ can be represented by

$$\begin{aligned} \mathbf{Q} &= [\mathbf{q}_l^1 \ \cdots \ \mathbf{q}_l^R] \in \mathbb{R}^{D_l \times R}, \\ \mathbf{Q} - \mathbf{U} &= [\mathbf{q}_l^1 - \mathbf{u} \ \cdots \ \mathbf{q}_l^R - \mathbf{u}] \in \mathbb{R}^{D_l \times R}, \\ \Sigma &= (\mathbf{Q} - \mathbf{U})(\mathbf{Q} - \mathbf{U})^\top / (R - 1) \in \mathbb{R}^{D_l \times D_l}, \\ \Lambda &= \mathbf{V}^\top \Sigma \mathbf{V} \in \mathbb{R}^{B \times B}, \end{aligned} \quad (4)$$

where, $\mathbf{u} = [u_1 \ \cdots \ u_{D_l}]^\top$. u_d is the average value of the d -th dimension of the vector $\mathbf{q}_l$. B is the number of dimensions after dimensionality reduction by PCA. Additionally,

$\Lambda = \text{diag}([\lambda_1 \cdots \lambda_B]^\top)$, where $\lambda_1, \dots, \lambda_B$ are the eigenvalues of Σ sorted in descending order. Let us denote the eigenvectors corresponding to these eigenvalues by $\mathbf{v}_1, \dots, \mathbf{v}_B$. The matrix $\mathbf{V} = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_B] \in \mathbb{R}^{D_l \times B}$ is formed by these eigenvectors. In this case, the process of reducing the dimension of the observation matrix $\mathbf{Q}$ to B dimensions is expressed as $\mathbf{V}^\top(\mathbf{Q} - \mathbf{U})$. That is, the dimensionality reduction of the vector $\mathbf{q}_l$ generated by an input image and the CNN is expressed as

$$\mathbf{p} = p(\mathbf{q}_l) = \mathbf{V}^\top(\mathbf{q}_l - \mathbf{u}), \quad p: \mathbb{R}^{D_l} \mapsto \mathbb{R}^B, \quad B \leq D_l. \quad (5)$$

C. SVM layer

As shown in ‘‘SVM Classification’’ in Figure 1, we assume the use of an SVM taking the PCA feature vector $\mathbf{p}$ as input.

The SVM predicted class $s(\mathbf{p})$ is defined as

$$s(\mathbf{p}) = \text{sign}(a(\mathbf{p})), \quad a(\mathbf{p}) = \sum_{i=1}^I \alpha_i c_i K(\mathbf{p}_i, \mathbf{p}) + g, \quad (6)$$

where $\text{sign}(\cdot)$ is a function that returns 1 if the input value is greater than 0, and returns -1 if the input value is less than 0. Moreover, g is the bias, I is the number of support vectors, i is the index of a support vector, $\mathbf{p}_i$ is the i -th support vector, c_i is the i -th actual class, α_i is the Lagrange multiplier corresponding to the i -th support vector. Additionally, $K(\mathbf{p}_i, \mathbf{p})$ denotes the kernel function. In this study, we use the linear and RBF kernels, which are expressed as

$$K(\mathbf{p}_i, \mathbf{p}) = \begin{cases} \mathbf{p}_i^\top \mathbf{p}, & \text{‘‘Linear kernel’’} \\ \exp(-\gamma \|\mathbf{p}_i - \mathbf{p}\|_2^2), & \text{‘‘RBF kernel’’} \end{cases},$$

$$\|\mathbf{p}_i - \mathbf{p}\|_2^2 = (\mathbf{p}_i - \mathbf{p})^\top (\mathbf{p}_i - \mathbf{p}), \quad \gamma > 0. \quad (7)$$

D. Conventional-Grad-CAM

Next, we describe ‘‘Conventional-Grad-CAM’’ as shown in Figure 1. This is a popular CAM method proposed by Selvaraju et al. [8]. Using a block matrix representation, the Grad-CAM for class c is defined as

$$\begin{aligned} \mathbf{G}_c &= \text{Relu}(\mathbf{a}_c^\top \mathbf{F}) \in [0, \infty)^{M \times N}, \\ \mathbf{a}_c &= \begin{bmatrix} a_c^1 \\ \vdots \\ a_c^T \end{bmatrix} \in \mathbb{R}^T, \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}^1 \\ \vdots \\ \mathbf{F}^T \end{bmatrix} \in \mathbb{R}^{MT \times N}, \end{aligned} \quad (8)$$

where $\mathbf{a}_c^\top \mathbf{F}$ appearing in $\mathbf{a}_c^\top \mathbf{F}$ denotes the scalar multiplication of the matrix $\mathbf{F}^t$ by $\mathbf{a}_c^\top$. Herein, a_c^t represents the weight of the t -th feature map $\mathbf{F}^t$ for class c , and is defined as

$$a_c^t = \sum_{m=1}^M \sum_{n=1}^N \frac{\partial y_c}{\partial f_{m,n}^t} = \sum_{i=1}^{MN} \frac{\partial y_c}{\partial x_i^t}. \quad (9)$$

Note that Equation 1 is used for this transformation. As shown in Table 1, the conventional Grad-CAM described here has been applied in various studies.

IV. PROPOSED METHOD

In this section, we explain the details of PCA- and SVM-Grad-CAM by extending the existing Grad-CAM method described in Section III. Subsections IV-A and IV-B correspond to PCA-Grad-CAM and SVM-Grad-CAM, respectively.

TABLE II
CLOSED-FORM JACOBIANS FOR PCA- AND SVM-GRAD-CAMS, ONE OF OUR KEY CONTRIBUTIONS. PROOF $\rightarrow$ EQUATIONS 13, 15, 20, 22, 23, 24, 25, AND 26.

Act	Kernel	Closed-form (PCA-Grad-CAM)
Sigm*	-	$\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} = \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}((1 - f(\delta_{l-j+1})) \odot f(\delta_{l-j+1})) \mathbf{W}_{l-j+1} \right) \text{diag}((1 - f(\delta_1)) \odot f(\delta_1)) \mathbf{W}_1^t$
Relu	-	$\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} = \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}(\mathbf{1}_{\delta_{l-j+1}}) \mathbf{W}_{l-j+1} \right) \text{diag}(\mathbf{1}_{\delta_1}) \mathbf{W}_1^t$
Act	Kernel	Closed-form (SVM-Grad-CAM)
Sigm*	Lin	$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = \left(\sum_{i=1}^I \alpha_i c_i \mathbf{p}_i^\top \right) \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}((1 - f(\delta_{l-j+1})) \odot f(\delta_{l-j+1})) \mathbf{W}_{l-j+1} \right) \text{diag}((1 - f(\delta_1)) \odot f(\delta_1)) \mathbf{W}_1^t$
Sigm*	RBF	$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = 2\gamma \left(\sum_{i=1}^I \alpha_i c_i \exp(-\gamma \ \mathbf{p}_i - \mathbf{p}\ _2^2) (\mathbf{p}_i - \mathbf{p})^\top \right) \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}((1 - f(\delta_{l-j+1})) \odot f(\delta_{l-j+1})) \mathbf{W}_{l-j+1} \right) \text{diag}((1 - f(\delta_1)) \odot f(\delta_1)) \mathbf{W}_1^t$
Relu	Lin	$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = \left(\sum_{i=1}^I \alpha_i c_i \mathbf{p}_i^\top \right) \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}(\mathbf{1}_{\delta_{l-j+1}}) \mathbf{W}_{l-j+1} \right) \text{diag}(\mathbf{1}_{\delta_1}) \mathbf{W}_1^t$
Relu	RBF	$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = 2\gamma \left(\sum_{i=1}^I \alpha_i c_i \exp(-\gamma \ \mathbf{p}_i - \mathbf{p}\ _2^2) (\mathbf{p}_i - \mathbf{p})^\top \right) \mathbf{V}^\top \left(\prod_{j=1}^{l-1} \text{diag}(\mathbf{1}_{\delta_{l-j+1}}) \mathbf{W}_{l-j+1} \right) \text{diag}(\mathbf{1}_{\delta_1}) \mathbf{W}_1^t$

Act: CNN activation function, Kernel: SVM kernel function, Sigm: sigmoid function, Lin: linear kernel, RBF: RBF kernel.
*: $f(z) = (1 + \exp(-z))^{-1}$.

A. PCA-Grad-CAM

The plus-minus sign of the partial derivative $\partial y_c / \partial x_i^t$ appearing in Equation 9, which is required for conventional Grad-CAM, indicates the direction in which the estimated class score changes due to the local effect of a feature map pixel. Specifically, a positive sign indicates that the pixel has a positive effect on the estimated class, whereas a negative sign indicates a negative effect. The objective of conventional Grad-CAM is to clearly highlight the focus region when a CNN predicts a class. Therefore, only the positive effects are visualized using the ReLU function, as shown in Equation 8. In contrast, both the positive and negative signs of a PCA feature vector are important. Therefore, visualizing only the positive focus regions is insufficient for PCA-Grad-CAM; that is, visualizing the negative focus regions is also important. For this reason, PCA-Grad-CAM should include both positive and negative focus regions.

Specifically, we define the PCA-Grad-CAM of the b -th principal component as

$$\mathbf{P}_b = \mathbf{e}_b^\top \mathbf{F} \in (-\infty, \infty)^{M \times N},$$

$$\mathbf{e}_b = [\mathbf{e}_b^1 \quad \dots \quad \mathbf{e}_b^T]^\top \in \mathbb{R}^T, \quad \mathbf{e}_b^t = \sum_{i=1}^{MN} \frac{\partial p_b}{\partial x_i^t}, \quad (10)$$

where $\mathbf{e}_b^t$ indicates the weight of the t -th feature map $\mathbf{F}^t$ for the b -th principal component. When regions in $\mathbf{P}_b$ have positive values, the corresponding pixels have a positive effect on the b -th principal component, whereas regions with negative values have a negative effect. If a heat map includes both positive and negative effects, understanding the focus regions of a CNN becomes difficult. Therefore, we split the PCA-Grad-CAM $\mathbf{P}_b$ into two separate heat maps representing the positive and negative effects. Specifically, $\mathbf{P}_b$ is split into

$$\mathbf{P}_b^\oplus = \text{Relu}(\mathbf{P}_b), \quad \mathbf{P}_b^\ominus = \text{Relu}(-\mathbf{P}_b). \quad (11)$$

$\mathbf{P}_b^\oplus$ visualizes the regions having a positive gradient effect on the b -th principal component p_b , whereas $\mathbf{P}_b^\ominus$ visualizes the regions having a negative gradient effect on the same component. Since it is preferable to compare the heat maps of

$\mathbf{P}_b^\oplus$ and $\mathbf{P}_b^\ominus$, we adopt

$$\nu_b = \max_{\omega \in \{\oplus, \ominus\}} \left\{ \max_{m \in \mathbb{N}_{\leq M}, n \in \mathbb{N}_{\leq N}} (\mathbf{P}_b^\omega)_{m,n} \right\} \quad (12)$$

as the maximum value of the color bar when visualizing these heat maps. ν_b is defined as the larger of the maximum values of $\mathbf{P}_b^\oplus$ and $\mathbf{P}_b^\ominus$.

Figure 2 shows the PCA-Grad-CAMs obtained from a CNN trained on the CIFAR-10 dataset [66]. The details of this model are described in Section VI. $\mathbf{P}_1^\oplus$ shown in Figure 2(a) is the PCA-Grad-CAM representing the positive gradient effect on the first principal component, indicating that this component slightly focuses on an airplane. In contrast, $\mathbf{P}_1^\ominus$ shown in the same figure represents the negative gradient effect on the first principal component, indicating that this component strongly focuses on the sky. As in the comparison shown here, using ν_b as the maximum of the color bar allows comparison of $\mathbf{P}_b^\oplus$ and $\mathbf{P}_b^\ominus$. In summary, p_1 , the first principal component of the PCA feature vector, is decreased by the sky region. Next, we focus on the second principal component, as shown in Figure 2 (a). $\mathbf{P}_2^\oplus$ indicates that the airplane regions are strongly activated. In contrast, $\mathbf{P}_2^\ominus$ has almost no activated regions. Therefore, the second principal component p_2 of the PCA feature vector can be interpreted as being increased by the airplane regions.

The above-mentioned tendencies of $\mathbf{P}_1^\oplus$, $\mathbf{P}_1^\ominus$, $\mathbf{P}_2^\oplus$, and $\mathbf{P}_2^\ominus$ are almost the same not only in Figure 2 (a) but also in the cases of (b) and (c). As illustrated in the aforementioned example, by examining the PCA-Grad-CAMs $\mathbf{P}_b^\oplus$ and $\mathbf{P}_b^\ominus$, we can identify the focus regions of each component of the PCA feature vector. It should be noted that Grad-CAM uses the partial derivatives as the weights of the feature maps. Therefore, even if $\mathbf{P}_b^\oplus$ is strongly activated, p_b is not always a positive value. Similarly, even if $\mathbf{P}_b^\ominus$ is strongly activated, p_b is not always a negative value.

As shown in Equation 10, $\partial p_b / \partial x_i^t$ is required for obtaining PCA-Grad-CAM. This variable can be expressed in matrix form as

$$\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} = \left[\frac{\partial p_b}{\partial x_i^t} \right] \in \mathbb{R}^{B \times MN}, \quad b \in \mathbb{N}_{\leq B}, \quad i \in \mathbb{N}_{\leq MN},$$

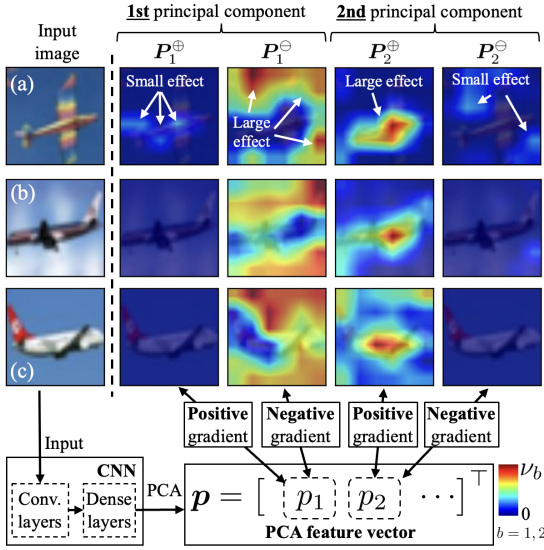


Fig. 2. PCA-Grad-CAM visualizations generated from a CNN trained on the CIFAR-10 dataset, as described in Section VI.

and it is called the Jacobian. By using this expression and Equations 2, 3, and 5, we obtain

$$\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} = \underbrace{\frac{\partial \mathbf{p}}{\partial \mathbf{q}_l}}_{\text{PCA}} \underbrace{\frac{\partial \mathbf{q}_l}{\partial \mathbf{q}_{l-1}} \dots \frac{\partial \mathbf{q}_2}{\partial \mathbf{q}_1}}_{\text{Dense}} \underbrace{\frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t}}_{\text{Flat}}, \quad l < L. \quad (13)$$

The meanings of the left-hand side and the right-hand side of this equation are shown in Figure 3. This figure assumes that a PCA layer is implemented in the l -th dense layer ($l = 2$) of a CNN, as shown in Figure 1 and Equation 3. The terms $\partial \mathbf{p} / \partial \mathbf{q}_2$, $\mathbf{q}_2 / \mathbf{q}_1$, and $\partial \mathbf{q}_1 / \partial \mathbf{x}^t$ appearing on the right-hand side of Equation 13 represent the Jacobians of directly connected dense layers. On the other hand, $\partial \mathbf{p} / \partial \mathbf{x}^t$, the left-hand side of Equation 13, represents the Jacobian expanded using the gradient chain rule. As shown in the differential graph of $\partial \mathbf{p} / \partial \mathbf{x}^t$ in Figure 3, we can clearly see the gradient effect of each pixel in the t -th feature map on the PCA feature vector $\mathbf{p}$. By summing over all paths from $\mathbf{x}^t$ to $\mathbf{p}$, the weight of the PCA-Grad-CAM, e_b^t , is defined as shown in Equation 10. The closed form of $\partial \mathbf{p} / \partial \mathbf{x}^t$ is shown in Table II. The formulas depend on the activation function used in the CNN. We provide solutions for the Sigmoid and Relu activation functions. The exact proof is presented in Subsection IV-C.

B. SVM-Grad-CAM

Here, we explain a novel method, ‘‘SVM-Grad-CAM,’’ as illustrated in Figure 1. Referring to the conventional Grad-CAM defined in Equations 8 and 9, we propose SVM-Grad-CAM as

$$\mathbf{S} = \text{Relu}(\mathbf{s}^\top \mathbf{F}) \in [0, \infty)^{M \times N}, \quad \mathbf{s} = [\mathbf{s}^1 \quad \dots \quad \mathbf{s}^T]^\top \in \mathbb{R}^T, \quad \mathbf{s}^t = \sum_{i=1}^{MN} \frac{\partial a(\mathbf{p})}{\partial \mathbf{x}_i^t}. \quad (14)$$

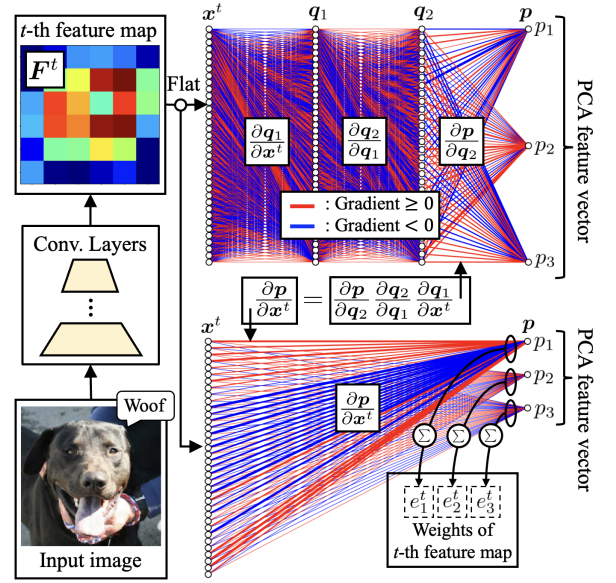


Fig. 3. Visualized Jacobian computed from a CNN trained on the Dog vs. Cat dataset, as described in Section VI. The red and blue paths represent positive and negative gradients, respectively. The width of each path represents the absolute value of the gradient.

The gradients required to compute $\mathbf{s}^t$ can be summarized as

$$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = \left[\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}_i^t} \right] \in \mathbb{R}^{1 \times MN}, \quad i \in \mathbb{N}_{\leq MN}.$$

By using this expression and Equations 2, 3, 5, and 6, we can obtain

$$\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} = \underbrace{\frac{\partial a(\mathbf{p})}{\partial \mathbf{p}}}_{\text{SVM}} \underbrace{\frac{\partial \mathbf{p}}{\partial \mathbf{q}_l}}_{\text{PCA}} \underbrace{\frac{\partial \mathbf{q}_l}{\partial \mathbf{q}_{l-1}} \dots \frac{\partial \mathbf{q}_2}{\partial \mathbf{q}_1}}_{\text{Dense}} \underbrace{\frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t}}_{\text{Flat}}, \quad l < L. \quad (15)$$

This closed-form depends on the CNN activation function (Sigmoid or ReLU) and the SVM kernel function (Linear or RBF), and it is shown in Table II. The exact proof is provided in Subsection IV-C.

C. Closed-form Jacobians

To obtain PCA- and SVM-Grad-CAM, the closed-forms of $\partial \mathbf{q}_l / \partial \mathbf{q}_{l-1}$, $\partial \mathbf{q}_l / \partial \mathbf{x}^t$, $\partial \mathbf{p} / \partial \mathbf{q}_l$, and $\partial a(\mathbf{p}) / \partial \mathbf{x}^t$ appearing in Equations 13 and 15 are required. These are shown in Table II. In this subsection, we present the exact proofs of these formulas.

1) Jacobian from the dense layer to the next dense layer:

When we represent an affine transform $\mathbf{W}_l \mathbf{q}_{l-1} + \beta_l$ as

$$\delta_l = \mathbf{W}_l \mathbf{q}_{l-1} + \beta_l, \quad l > 1, \quad (16)$$

we can represent $\mathbf{q}_l = f(\delta_l)$ as the transformation of the l -th dense layer, as shown in Figure 1 and Equation 3. Then, the Jacobian $\partial \mathbf{q}_l / \partial \mathbf{q}_{l-1}$ can be expressed as

$$\frac{\partial \mathbf{q}_l}{\partial \mathbf{q}_{l-1}} = \frac{\partial f(\delta_l)}{\partial \delta_l} \frac{\partial \delta_l}{\partial \mathbf{q}_{l-1}} = \frac{\partial f(\delta_l)}{\partial \delta_l} \mathbf{W}_l \in \mathbb{R}^{D_l \times D_{l-1}}. \quad (17)$$

That is, the Jacobian depends on the activation function f . When we represent i -th dimension of δ_l as $(\delta_l)_i = \delta_{li}$ and

adopt the Sigmoid function as the activation function, we obtain

$$\begin{aligned} f(z) &= (1 + \exp(-z))^{-1} \Rightarrow \\ \left(\frac{\partial f(\delta_l)}{\partial \delta_l} \right)_{ij} &= \frac{\partial(1 + \exp(-\delta_{li}))^{-1}}{\partial \delta_{li}} \\ &= \begin{cases} (1 - f(\delta_{li}))f(\delta_{li}), & i = j \\ 0, & i \neq j \end{cases} \\ \therefore \frac{\partial f(\delta_l)}{\partial \delta_l} &= \text{diag}((1 - f(\delta_l)) \odot f(\delta_l)) \in \mathbb{R}^{D_l \times D_l}. \end{aligned} \quad (18)$$

Here, $\odot$ denotes Hadamard product, and $\text{diag}(z)$ denotes the diagonal matrix of the vector z . When we adopt the Relu function as the activation, we obtain

$$\begin{aligned} f(z) &= \text{Relu}(z) \Rightarrow \\ \left(\frac{\partial f(\delta_l)}{\partial \delta_l} \right)_{ij} &= \frac{\partial \text{Relu}(\delta_{li})}{\partial \delta_{li}} = \begin{cases} 1, & i = j \wedge \delta_{li} > 0 \\ 0, & \text{otherwise} \end{cases} \\ \therefore \frac{\partial f(\delta_l)}{\partial \delta_l} &= \text{diag}(\mathbf{1}_{\delta_l}) \in \{0, 1\}^{D_l \times D_l}, \quad \mathbf{1}_{\delta_l} \in \{0, 1\}^{D_l}, \end{aligned} \quad (19)$$

where

$$(\mathbf{1}_{\delta_l})_i = \begin{cases} 1, & \delta_{li} > 0 \\ 0, & \text{otherwise} \end{cases}.$$

Therefore, by using Equations 17, 18, and 19, the closed-form Jacobian can be expressed as

$$\frac{\partial q_l}{\partial q_{l-1}} = \begin{cases} \text{diag}((1 - f(\delta_l)) \odot f(\delta_l)) \mathbf{W}_l, & f : \text{“Sigmoid”} \\ \text{diag}(\mathbf{1}_{\delta_l}) \mathbf{W}_l, & f : \text{“Relu”} \end{cases}. \quad (20)$$

2) *Jacobian from the flatten layer to the dense layer:* Here, we describe the Jacobian $\partial q_1 / \partial x^t$. The output $q_1 \in \mathbb{R}^{D_1}$ with respect to the input x is computed as $q_1 = f(\mathbf{W}_1 x + \beta_1)$. Similar to Equation 16, we adopt

$$\delta_1 = \mathbf{W}_1 x + \beta_1 \in \mathbb{R}^{D_1}. \quad (21)$$

Since x^t is used in $\partial q_1 / \partial x^t$ instead of x , unlike in Equation 16, we express the weight matrix $\mathbf{W}_1$ in terms of

$$\mathbf{W}_1 = [\mathbf{W}_1^1 \quad \dots \quad \mathbf{W}_1^T] \in \mathbb{R}^{D_1 \times MNT}, \quad \mathbf{W}_1^t \in \mathbb{R}^{D_1 \times MN}.$$

By adopting this expression, $\mathbf{W}_1^t$ corresponds to x^t in Equations 1 and 2. Specifically, δ_1 can be expressed as

$$\delta_1 = \sum_{t=1}^T \mathbf{W}_1^t x^t + \beta_1 \in \mathbb{R}^{D_1}.$$

Therefore, we obtain

$$\frac{\partial q_1}{\partial x^t} = \frac{\partial f(\delta_1)}{\partial \delta_1} \frac{\partial \delta_1}{\partial x^t} = \frac{\partial f(\delta_1)}{\partial \delta_1} \mathbf{W}_1^t \in \mathbb{R}^{D_1 \times MN}.$$

And, by using Equations 18 and 19 we obtain

$$\frac{\partial q_1}{\partial x^t} = \begin{cases} \text{diag}((1 - f(\delta_1)) \odot f(\delta_1)) \mathbf{W}_1^t, & f : \text{“Sigmoid”} \\ \text{diag}(\mathbf{1}_{\delta_1}) \mathbf{W}_1^t, & f : \text{“Relu”} \end{cases}. \quad (22)$$

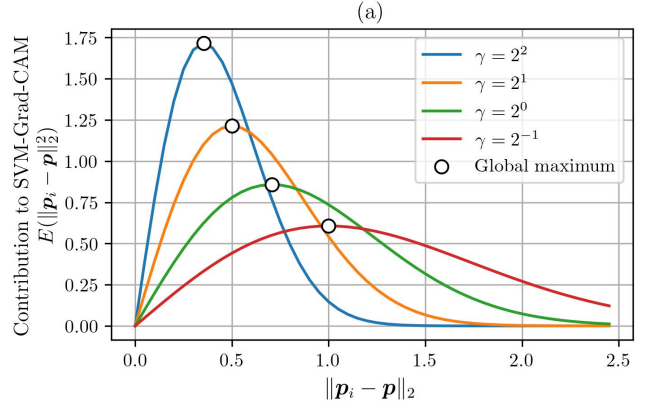


Fig. 4. Effect of the distance between the PCA feature vector p and the i -th support vector p_i , $\|p_i - p\|_2$, on $E(\|p_i - p\|_2)$. $E(\|p_i - p\|_2)$ represents the degree of effect on the SVM-Grad-CAM S and is defined in Equation 27. From Theorem 2, the distance that maximizes $E(\|p_i - p\|_2)$ is $\|p_i - p\|_2 = (2\gamma)^{-1/2}$, and the $\circ$ in the figure represents these locations.

3) *On the PCA and SVM layers Jacobian:* From Equation 5, the Jacobian of the PCA layer, $\partial p / \partial q_l$, is expressed by

$$\frac{\partial p}{\partial q_l} = \frac{\partial V^T(q_l - u)}{\partial q_l} = V^T \in \mathbb{R}^{B \times D_l}. \quad (23)$$

In the case of the SVM layer, from Equation 6, $\partial a(p) / \partial p$ can be expressed by

$$\frac{\partial a(p)}{\partial p} = \sum_{i=1}^I \alpha_i c_i \frac{\partial K(p_i, p)}{\partial p} \in \mathbb{R}^{1 \times B}. \quad (24)$$

That is, the gradient of the kernel function $K(p_i, p)$ is required. From Equation 7, when adopting the linear kernel, the gradient can be expressed by

$$K(p_i, p) = p_i^T p \Rightarrow \frac{\partial K(p_i, p)}{\partial p} = p_i^T \in \mathbb{R}^{1 \times B}. \quad (25)$$

On the other hand, from Equation 7, when adopting the RBF kernel, the gradient can be expressed as

$$\begin{aligned} K(p_i, p) &= \exp(-\gamma \|p_i - p\|_2^2) \Rightarrow \\ \frac{\partial K(p_i, p)}{\partial p} &= \frac{\partial \exp(-\gamma \|p_i - p\|_2^2)}{\partial (-\gamma \|p_i - p\|_2^2)} \frac{\partial (-\gamma \|p_i - p\|_2^2)}{\partial p} \\ &= 2\gamma \exp(-\gamma \|p_i - p\|_2^2) (p_i - p)^T \in \mathbb{R}^{1 \times B}, \end{aligned} \quad (26)$$

where γ is the hyperparameter of the RBF kernel and relates to the decision boundary of the classification.

From these closed-forms, we can understand the characteristics of the SVM-Grad-CAM. From Equation 25, when adopting the linear kernel, we can interpret that only the locations of the support vectors p_i ($\forall i \in \mathbb{N}_{\leq I}$) affect $\partial K(p_i, p) / \partial p$. On the other hand, from Equation 26, when adopting the RBF kernel, both p and p_i influence $\partial K(p_i, p) / \partial p$. As shown in Equations 14, 15, and 24, $\partial K(p_i, p) / \partial p$ directly affects the SVM-Grad-CAM S . Therefore, we define the degree of effect of the support vector p_i on SVM-Grad-CAM as the L2 norm of $\partial K(p_i, p) / \partial p$, as defined in Equation 26. That is,

$$\begin{aligned} E(p_i, p) &= \|\partial K(p_i, p) / \partial p\|_2 \\ &= 2\gamma \exp(-\gamma \|p_i - p\|_2^2) \|p_i - p\|_2. \end{aligned} \quad (27)$$

When E has a large value, we interpret that the support vector $\mathbf{p}_i$ strongly affects the SVM-Grad-CAM $\mathcal{S}$. Since $E(\mathbf{p}_i, \mathbf{p})$ can be regarded as a single-variable function of $\|\mathbf{p}_i - \mathbf{p}\|_2$, we express

$$E(\|\mathbf{p}_i - \mathbf{p}\|_2) = E(\mathbf{p}_i, \mathbf{p}). \quad (28)$$

This indicates that the distance between the support vector $\mathbf{p}_i$ and the observation PCA vector $\mathbf{p}$ affects the SVM-Grad-CAM $\mathcal{S}$. The shape of $E(\mathbf{p}_i, \mathbf{p})$ is shown in Figure 4 and indicates that when the distance is too large, $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ approaches zero. Moreover, if the distance approaches zero, $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ also approaches zero. This visual observation can be proven as the following theorem.

Theorem 1:

$$\lim_{\|\mathbf{p}_i - \mathbf{p}\|_2 \rightarrow +\infty} E(\|\mathbf{p}_i - \mathbf{p}\|_2) = 0, \quad (29)$$

$$\lim_{\|\mathbf{p}_i - \mathbf{p}\|_2 \rightarrow 0} E(\|\mathbf{p}_i - \mathbf{p}\|_2) = 0 \quad (30)$$

Proof: Equation 29 is omitted since it is trivial. Since Equation 30 yields an indeterminate form, we represent Equations 27 and 28 as

$$E(\|\mathbf{p}_i - \mathbf{p}\|_2) = \frac{\hat{E}(\|\mathbf{p}_i - \mathbf{p}\|_2)}{\tilde{E}(\|\mathbf{p}_i - \mathbf{p}\|_2)} = \frac{2\gamma\|\mathbf{p}_i - \mathbf{p}\|_2}{\exp(\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2)}.$$

Using Generalized L'Hôpital's rule [67], we obtain

$$\begin{aligned} & \lim_{\|\mathbf{p}_i - \mathbf{p}\|_2 \rightarrow 0} E(\|\mathbf{p}_i - \mathbf{p}\|_2) \\ &= \lim_{\|\mathbf{p}_i - \mathbf{p}\|_2 \rightarrow 0} \frac{d\hat{E}(\|\mathbf{p}_i - \mathbf{p}\|_2)/d\|\mathbf{p}_i - \mathbf{p}\|_2}{d\tilde{E}(\|\mathbf{p}_i - \mathbf{p}\|_2)/d\|\mathbf{p}_i - \mathbf{p}\|_2} \\ &= \lim_{\|\mathbf{p}_i - \mathbf{p}\|_2 \rightarrow 0} \frac{1}{\|\mathbf{p}_i - \mathbf{p}\|_2 \exp(\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2)} = 0. \end{aligned}$$

This theorem indicates that when the distance is either too short or too long, $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$, the effect on SVM-Grad-CAM $\mathcal{S}$, converges to zero. In other words, maintaining a moderate distance is important. Therefore, we consider the distance $\|\mathbf{p}_i - \mathbf{p}\|_2$ that maximizes $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$. This is explained in the following theorem.

Theorem 2:

$$\operatorname{argmax}_{\|\mathbf{p}_i - \mathbf{p}\|_2 \in [0, \infty)} E(\|\mathbf{p}_i - \mathbf{p}\|_2) = \frac{1}{\sqrt{2\gamma}} \quad (31)$$

$$\max_{\|\mathbf{p}_i - \mathbf{p}\|_2 \in [0, \infty)} E(\|\mathbf{p}_i - \mathbf{p}\|_2) = \sqrt{2\gamma} \exp\left(-\frac{1}{2}\right) \quad (32)$$

Proof: The first and second order derivatives of $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ with respect to the distance can be expressed as

$$\begin{aligned} & \frac{dE(\|\mathbf{p}_i - \mathbf{p}\|_2)}{d\|\mathbf{p}_i - \mathbf{p}\|_2} = 2\gamma(1 - 2\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2) \exp(-\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2), \\ & \frac{d^2E(\|\mathbf{p}_i - \mathbf{p}\|_2)}{d(\|\mathbf{p}_i - \mathbf{p}\|_2)^2} = \\ & -4\gamma^2\|\mathbf{p}_i - \mathbf{p}\|_2(3 - 2\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2) \exp(-\gamma\|\mathbf{p}_i - \mathbf{p}\|_2^2). \end{aligned}$$

When the distance is $1/\sqrt{2\gamma}$, we obtain

$$\begin{aligned} & \left. \frac{dE(\|\mathbf{p}_i - \mathbf{p}\|_2)}{d\|\mathbf{p}_i - \mathbf{p}\|_2} \right|_{\|\mathbf{p}_i - \mathbf{p}\|_2 = \frac{1}{\sqrt{2\gamma}}} = 0, \\ & \left. \frac{d^2E(\|\mathbf{p}_i - \mathbf{p}\|_2)}{d(\|\mathbf{p}_i - \mathbf{p}\|_2)^2} \right|_{\|\mathbf{p}_i - \mathbf{p}\|_2 = \frac{1}{\sqrt{2\gamma}}} = -4\sqrt{2\gamma}^{\frac{3}{2}} \exp\left(-\frac{1}{2}\right) < 0. \end{aligned}$$

Therefore, $\|\mathbf{p}_i - \mathbf{p}\|_2 = 1/\sqrt{2\gamma}$ is a critical point of $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ and corresponds to a local maximum. Moreover, since

$$\left. \frac{dE(\|\mathbf{p}_i - \mathbf{p}\|_2)}{d\|\mathbf{p}_i - \mathbf{p}\|_2} \right\} \begin{cases} > 0, & 0 \leq \|\mathbf{p}_i - \mathbf{p}\|_2 < \frac{1}{\sqrt{2\gamma}} \\ < 0, & \|\mathbf{p}_i - \mathbf{p}\|_2 > \frac{1}{\sqrt{2\gamma}} \end{cases}$$

is satisfied, $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ is a monotonically increasing function when the distance is shorter than the critical point. In contrast, $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ is a monotonically decreasing function when the distance is longer than the critical point. Therefore, the critical point $\|\mathbf{p}_i - \mathbf{p}\|_2 = 1/\sqrt{2\gamma}$ corresponds to the global maximum of $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$, which means that Equation 31 is satisfied. Moreover, substituting it into $E(\|\mathbf{p}_i - \mathbf{p}\|_2)$ yields Equation 32. ■

This theorem indicates that when the distance between the i -th support vector $\mathbf{p}_i$ and the observed PCA feature vector $\mathbf{p}$ is $1/\sqrt{2\gamma}$, the effect of $\mathbf{p}_i$ on SVM-Grad-CAM $\mathcal{S}$ is maximized. ○ in Figure 4 marks this location. From Theorem 2 and Figure 4, we can understand the SVM hyperparameter γ directly affects the SVM-Grad-CAM $\mathcal{S}$. Specifically, when γ is small, the distance required to maximize the effect is also short.

V. ALGORITHM

Procedures for computing PCA- and SVM- Grad-CAM are shown in Algorithms 1 and 2. Algorithm 1 is the procedure for computing the Jacobians $\partial \mathbf{p} / \partial \mathbf{x}^t$ and $\partial a(\mathbf{p}) / \partial \mathbf{x}^t$, which are needed to compute the weights of feature maps. First, the feature maps $\{\mathbf{F}^1, \dots, \mathbf{F}^T\}$ corresponding to the input image X are extracted from the trained CNN Υ using the “GetFeatureMap” function. By flattening them, the input vector $\mathbf{x}$ is obtained. After that, the weights and biases, $\{(\mathbf{W}_1, \beta_1), \dots, (\mathbf{W}_l, \beta_l)\}$, for the affine transformations are obtained using the “GetParams” function. The above procedures correspond to lines 1–6 of Algorithm 1.

Next, the procedures related to the PCA transformation are carried out. Specifically, the input vector $\mathbf{x}$ is compressed into a B -dimensional PCA feature vector $\mathbf{p}$ using the covariance matrix Σ and the eigenmatrix $\mathbf{V}$, as shown in lines 7–14 of Algorithm 1. After that, the Jacobians of the SVM layer (lines 15–20 of Algorithm 1), the PCA layer (lines 21–22 of Algorithm 1), and the dense layers (lines 23–31 of Algorithm 1) are computed. From these, the closed-form Jacobians of $\{\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$ and $\{\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$ are computed using the gradient chain rule. Finally, these Jacobians and the feature maps $\mathbf{F}$ are returned. The above procedures are indicated in lines 32–43 of Algorithm 1.

Algorithm 2 shows the exact procedures for computing PCA- and SVM- Grad-CAM using the Jacobians. First, using $\{\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$ returned from Algorithm 1, the PCA-Grad-CAMs $\mathbf{P}_b^\oplus$ and $\mathbf{P}_b^\ominus$ corresponding to the b -th principal component are computed, as shown in lines 1–9 of Algorithm

Algorithm 1 Procedure for computing Jacobians

Input: The input image: X , the CNN model: Υ , the activation: $f \in \{\text{"Sigmoid"}, \text{"Relu"}\}$, the layer number where PCA is applied: l , the number of principal components: B , the covariance matrix: Σ , the average vector: $\mathbf{u}$, the kernel function: $k \in \{\text{"Linear"}, \text{"RBF"}\}$, the trained SVM parameters $\{(\alpha_i, c_i, \mathbf{p}_i) \mid i \in \mathbb{N}_{\leq I}\}$ and γ

Output: $\{\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, $\{\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, $\mathbf{F}$

- 1: **###** Feature extraction from the input image X **###**
- 2: $\{\mathbf{F}^1, \dots, \mathbf{F}^T\} \leftarrow \text{GetFeatureMap}(\Upsilon, X)$
- 3: $\{\mathbf{x}^1, \dots, \mathbf{x}^T\} \leftarrow \text{Flatten}(\{\mathbf{F}^1, \dots, \mathbf{F}^T\})$, Eq.1
- 4: $\mathbf{x} \leftarrow [(\mathbf{x}^1)^\top \dots (\mathbf{x}^T)^\top]^\top$, Eq. 2
- 5: $\mathbf{F} \leftarrow [(\mathbf{F}^1)^\top \dots (\mathbf{F}^T)^\top]^\top$, Eq. 8
- 6: $\{(\mathbf{W}_1, \beta_1), \dots, (\mathbf{W}_l, \beta_l)\} \leftarrow \text{GetParams}(\Upsilon)$, Eq.3
- 7: **###** Get PCA vector $\mathbf{p}$ **###**
- 8: $\{\mathbf{v}_1, \dots, \mathbf{v}_B\} \leftarrow \text{GetEigenvector}(\Sigma, B)$, Eq.4
- 9: $\mathbf{V} \leftarrow [\mathbf{v}_1 \dots \mathbf{v}_B]$, Eq.4
- 10: $\mathbf{q}_1 \leftarrow f(\mathbf{W}_1 \mathbf{x} + \beta_1)$, Eq.3
- 11: **for** $j \leftarrow 2$ **to** l **do**
- 12: $\mathbf{q}_j \leftarrow f(\mathbf{W}_j \mathbf{q}_{j-1} + \beta_j)$, Eq.3
- 13: **end for**
- 14: $\mathbf{p} \leftarrow \mathbf{V}^\top (\mathbf{q}_l - \mathbf{u})$, Eq.5
- 15: **###** Jacobian for the SVM layer **###**
- 16: **if** $k = \text{"Linear"}$ **then**
- 17: $\frac{\partial a(\mathbf{p})}{\partial \mathbf{p}} \leftarrow \sum_{i=1}^I \alpha_i c_i \mathbf{p}_i^\top$, Eqs.24, 25
- 18: **else if** $k = \text{"RBF"}$ **then**
- 19: $\frac{\partial a(\mathbf{p})}{\partial \mathbf{p}} \leftarrow 2\gamma \sum_{i=1}^I \alpha_i c_i \exp(-\gamma \|\mathbf{p}_i - \mathbf{p}\|_2^2) (\mathbf{p}_i - \mathbf{p})^\top$, Eqs.24, 26
- 20: **end if**
- 21: **###** Jacobian for the PCA layer **###**
- 22: $\frac{\partial \mathbf{p}}{\partial \mathbf{q}_l} \leftarrow \mathbf{V}^\top$, Eq.23
- 23: **###** Jacobians for the dense layers **###**
- 24: **for** $j \leftarrow 2$ **to** l **do**
- 25: $\delta_j \leftarrow \mathbf{W}_j \mathbf{q}_{j-1} + \beta_j$, Eq.16
- 26: **if** $f = \text{"Sigmoid"}$ **then**
- 27: $\frac{\partial \mathbf{q}_j}{\partial \mathbf{q}_{j-1}} \leftarrow \text{diag}((1 - f(\delta_j)) \odot f(\delta_j)) \mathbf{W}_j$, Eq.20
- 28: **else if** $f = \text{"Relu"}$ **then**
- 29: $\frac{\partial \mathbf{q}_j}{\partial \mathbf{q}_{j-1}} \leftarrow \text{diag}(\mathbf{1}_{\delta_j}) \mathbf{W}_j$, Eq.20
- 30: **end if**
- 31: **end for**
- 32: **###** Jacobians for PCA- and SVM-Grad-CAM **###**
- 33: **for** $t \leftarrow 1$ **to** T **do**
- 34: $\delta_1 \leftarrow \mathbf{W}_1 \mathbf{x} + \beta_1$, Eq.21
- 35: **if** $f = \text{"Sigmoid"}$ **then**
- 36: $\frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t} \leftarrow \text{diag}((1 - f(\delta_1)) \odot f(\delta_1)) \mathbf{W}_1^t$, Eq.22
- 37: **else if** $f = \text{"Relu"}$ **then**
- 38: $\frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t} \leftarrow \text{diag}(\mathbf{1}_{\delta_1}) \mathbf{W}_1^t$, Eq.22
- 39: **end if**
- 40: $\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \leftarrow \frac{\partial \mathbf{p}}{\partial \mathbf{q}_l} \frac{\partial \mathbf{q}_l}{\partial \mathbf{q}_{l-1}} \dots \frac{\partial \mathbf{q}_2}{\partial \mathbf{q}_1} \frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t}$, Eq.13
- 41: $\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \leftarrow \frac{\partial a(\mathbf{p})}{\partial \mathbf{p}} \frac{\partial \mathbf{p}}{\partial \mathbf{q}_l} \frac{\partial \mathbf{q}_l}{\partial \mathbf{q}_{l-1}} \dots \frac{\partial \mathbf{q}_2}{\partial \mathbf{q}_1} \frac{\partial \mathbf{q}_1}{\partial \mathbf{x}^t}$, Eq.15
- 42: **end for**
- 43: **return** $\{\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, $\{\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, $\mathbf{F}$

Algorithm 2 Procedure for PCA- and SVM-Grad-CAM

Input: Feature maps matrix: $\mathbf{F}$, Jacobian for PCA-Grad-CAM: $\{\frac{\partial \mathbf{p}}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, Jacobian for SVM-Grad-CAM: $\{\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$, the number of principal components: B

Output: PCA-Grad-CAM: $\{\mathbf{P}_b^\oplus \mid b \in \mathbb{N}_{\leq B}\}$, $\{\mathbf{P}_b^\ominus \mid b \in \mathbb{N}_{\leq B}\}$, and SVM-Grad-CAM: $\mathbf{S}$

- 1: **###** Compute PCA-Grad-CAM **###**
- 2: **for** $b \leftarrow 1$ **to** B **do**
- 3: **for** $t \leftarrow 1$ **to** T **do**
- 4: $e_b^t \leftarrow \sum_{i=1}^{MN} \frac{\partial p_b}{\partial x_i^t}$, Eq.10
- 5: **end for**
- 6: $\mathbf{e}_b \leftarrow [e_b^1 \dots e_b^T]^\top$, Eq.10
- 7: $\mathbf{P}_b \leftarrow \mathbf{e}_b^\top \mathbf{F}$, Eq.10
- 8: $\mathbf{P}_b^\oplus \leftarrow \text{Relu}(\mathbf{P}_b)$, $\mathbf{P}_b^\ominus \leftarrow \text{Relu}(-\mathbf{P}_b)$, Eq.11
- 9: **end for**
- 10: **###** Compute SVM-Grad-CAM **###**
- 11: **for** $t \leftarrow 1$ **to** T **do**
- 12: $s^t \leftarrow \sum_{i=1}^{MN} \frac{\partial a(\mathbf{p})}{\partial x_i^t}$, Eq.14
- 13: **end for**
- 14: $\mathbf{s} \leftarrow [s^1 \dots s^T]^\top$, Eq.14
- 15: $\mathbf{S} \leftarrow \text{Relu}(\mathbf{s}^\top \mathbf{F})$, Eq.14
- 16: **return** $\{\mathbf{P}_b^\oplus \mid b \in \mathbb{N}_{\leq B}\}$, $\{\mathbf{P}_b^\ominus \mid b \in \mathbb{N}_{\leq B}\}$, $\mathbf{S}$

2. After that, using $\{\frac{\partial a(\mathbf{p})}{\partial \mathbf{x}^t} \mid t \in \mathbb{N}_{\leq T}\}$ returned from Algorithm 1, the SVM-Grad-CAM $\mathbf{S}$ is computed, as shown in lines 10–16 of Algorithm 2.

VI. EXPERIMENT

A. Outline

To verify the effectiveness of PCA- and SVM-Grad-CAMs, we conducted experiments using major datasets, including MNIST [68], Fashion-MNIST [69], Blood-MNIST [70], CIFAR-10 [66], Chest X-Ray [71], and Dog vs. Cat [72]. Note that Chest X-Ray dataset was created by Cohen et al. [71] and has been used in several studies [73]. The number of class labels, along with the sizes of the training and test datasets, are shown in Table III. The specific class label names are provided in the Appendix B. As the CNN, we adopted the ImageNet-based VGG16 [74] and implemented four dense layers immediately after the flatten layer. The dimensions of the first, second, and third dense layers are 40, 30, and 20, respectively. Since the fourth dense layer is used for classification, its dimension is the same as the number of class labels. The size of the input images was $200 \times 200 \times 3$ for all datasets. When adopting this input size with VGG16 [74], the Grad-CAM output size becomes 6×6 . Moreover, mini-batch learning with a batch size of 32 was performed. In the learning process, all convolutional layers were frozen, and only the dense layers were trained. An NVIDIA RTX A6000 (48 GB) was used for this experiment.

Stochastic Gradient Descent (SGD) [7] was used as the optimizer, cross-entropy was used as the loss function, and the learning rate was set to 10^{-3} . We allocated 20% of the

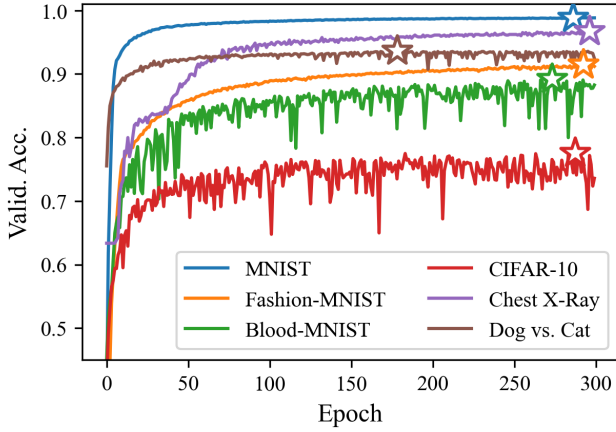


Fig. 5. Relationship between epoch and validation accuracies. A star marker ☆ represents a optimal epoch.

training data as validation data and searched for the optimal number of epochs that yielded the highest validation accuracy. The number of epochs was searched in the range from 1 to 300. The PCA layer was applied to the second dense layer ($l = 2$), and the dimension of the PCA feature vector was set to $B = 3$. In general, the dimensions are determined based on the contribution ratio computed from the eigenvalues. However, since the objective of this study is to evaluate PCA-Grad-CAM, the dimension of the PCA feature vector was set to the same value for all datasets. With a 3-dimensional PCA feature vector, the feature space can be visually inspected. Note that the PCA transformation matrix was constructed using only the training dataset; the test dataset was not used. As the SVM inserted after the PCA layer, we adopted two types: Linear SVM (Lin-SVM) and RBF-SVM. The hyperparameters of these SVMs were set as follows: the cost parameter was 1, and γ was 1. Since our objective was to verify SVM-Grad-CAM, no hyperparameter tuning was performed for the SVMs. For the above models, PCA-Grad-CAM and SVM-Grad-CAM were implemented.

B. Result and discussion

1) *Model development*: The relationship between epochs and validation accuracy is shown in Figure 5. This figure indicates that the validation accuracies have converged. The performances obtained by adopting the epoch that maximized the validation accuracy are shown in Table III. From this table,

TABLE III
DATASET INFORMATIONS AND ESTIMATION PERFORMANCES.

Dataset	Classes / $N_{\text{train}} / N_{\text{test}}$	Macro F1 score (training/test)		
		CNN	Lin-SVM	RBF-SVM
MNIST [68]	10/60,000/10,000	.993/.991	.986/.980	.997/.989
Fashion-MNIST [69]	10/60,000/10,000	.925/.903	.922/.897	.988/.943
Blood-MNIST [70]	8/11,959/3,421	.944/.877	.925/.873	.984/.909
CIFAR-10 [66]	10/50,000/10,000	.899/.767	.789/.735	.892/.794
Chest X-Ray [71]	3/5,144/1,288	.966/.956	.998/.975	.998/.975
Dog vs. Cat [72]	2/18,610/4,652	.967/.931	.995/.930	.995/.931

N_{train} : training samples, N_{test} : test samples

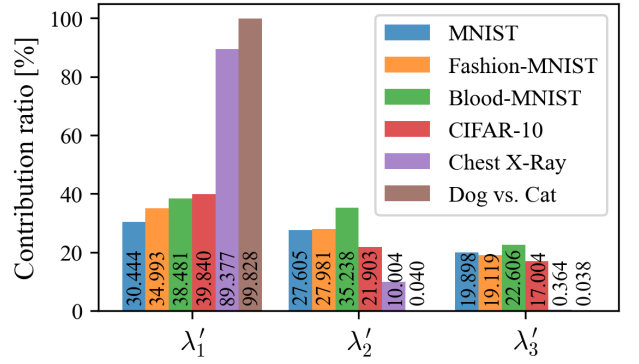


Fig. 6. Contribution ratios of the eigenvectors v_1 , v_2 , and v_3 . Here, λ'_b represents the normalized eigenvalues, defined as $\lambda'_b = 100 \times \lambda_b / \sum_b \lambda_b$.

it can be seen that the models exhibit good generalization performance. Therefore, we decided to use these CNNs for the experiments.

Next, the contribution ratios of the three principal components used in PCA are shown in Figure 6. These values are normalized eigenvalues, calculated as $\lambda'_b = 100 \times \lambda_b / \sum_b \lambda_b$. Note that λ_b denotes the b -th eigenvalue. Figure 6 shows that for models trained on the MNIST, Fashion-MNIST, Blood-MNIST, and CIFAR-10 datasets, the first to third principal components have relatively high contribution ratios. In contrast, for models trained on the Chest X-Ray and Dog vs. Cat datasets, only the first principal component has a high contribution ratio. The three-dimensional PCA feature spaces of all the models are shown in Figure 7. This figure indicates that the scatter plots for each class are somewhat separated. However, in the Dog vs. Cat model, PCA features such as p_2 and p_3 with low contribution ratios do not seem to improve classification performance. Therefore, we interpret that PCA features with large contribution ratios are important. Note that the total contribution ratios of the 1–3 principal components are greater than 75% for all the models.

Previous studies have shown that incorporating a PCA layer into a CNN can improve its classification performance [16] [17] [18] [19] [20] [21]. Figure 7 also demonstrates the effectiveness of PCA. However, we cannot know the focus regions of each principal component. PCA-Grad-CAM is a method that reveals this, and the results of its application are described in the next section.

2) *Visualization*: Next, the proposed methods, PCA-Grad-CAM and SVM-Grad-CAM, were computed using Algorithms 1 and 2. For comparison, the existing method, conventional Grad-CAM [8], was also computed. Figure 8 shows the cases in which the contribution ratios of the 1–3 principal components are relatively high, namely, the CNNs trained on the MNIST, Fashion-MNIST, Blood-MNIST, and CIFAR-10 datasets. In many cases, the focus regions of the SVM-Grad-CAM are similar to those of the conventional Grad-CAM. In contrast, for the “Trouser” class of Fashion-MNIST and the “Neutrophil” class of Blood-MNIST, the conventional Grad-CAM and the SVM-Grad-CAM produce different results. The conventional Grad-CAM was computed from a standard CNN,

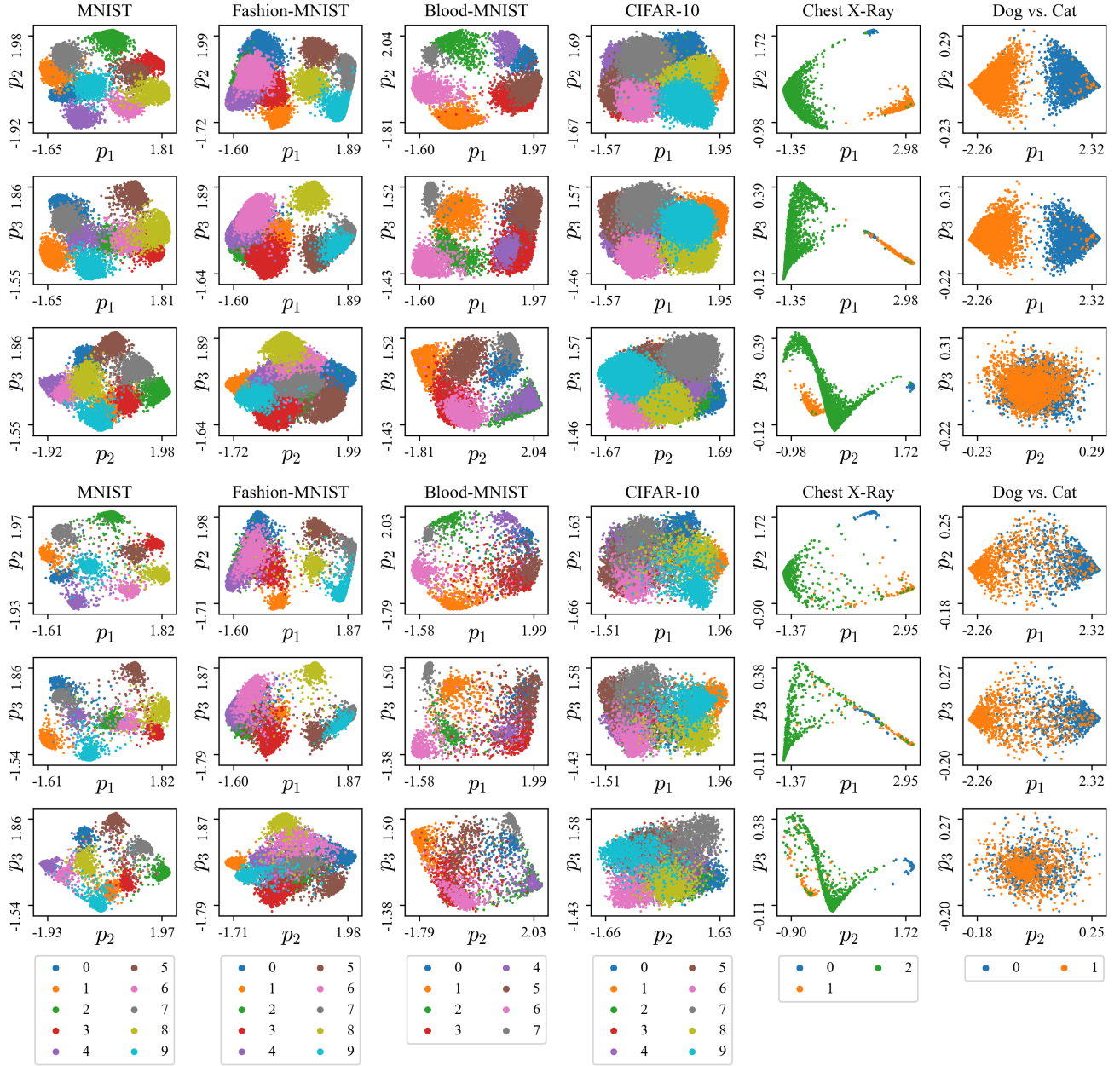


Fig. 7. Three-dimensional PCA feature spaces consisting of $\mathbf{p} = [p_1 \ p_2 \ p_3]^\top$. The meanings of the labels from 0 to 9 are described in the Appendix B. The top and bottom figures are scatter plots corresponding to the training and test data, respectively.

whereas the SVM-Grad-CAM was computed from an SVM embedded within the CNN. Therefore, it is natural that some results are similar while others are different.

Next, we examine the results of the PCA-Grad-CAM. $P_b^\oplus$ and $P_b^\ominus$ represent the Grad-CAMs for the b -th principal component p_b . As a general trend, we observe that the focus regions differ across the principal components. For example, for the class “4” in MNIST, $P_1^\oplus$ focuses on the right side, $P_2^\ominus$ on the top, and $P_3^\ominus$ on both the top and the background. For the class “9”, $P_1^\oplus$ focuses on the left and bottom, $P_2^\ominus$ on the center, and $P_3^\ominus$ on the top and background.

For the CIFAR-10 dataset, $P_1^\ominus$ focuses on the background, while $P_2^\oplus$ focuses on the airplane and the car. $P_3^\ominus$ has almost

all pixels activated. This indicates that the gradients of p_3 with respect to many feature maps were negative. Similarly, in the cases of Fashion-MNIST and Blood-MNIST, the focus regions differed among the principal components. This indicates that the 1–3 principal components extract features from different regions. Note that there are cases where PCA-Grad-CAM and SVM-Grad-CAM are similar. This is probably because the SVM performed classification using only that principal component. For example, in the MNIST class “4”, the SVM-Grad-CAM is very similar to $P_2^\ominus$. This suggests that the second dimension of the PCA feature vector, p_2 , likely contributes to the classification.

Next, in Figure 9, we show the results for cases where only

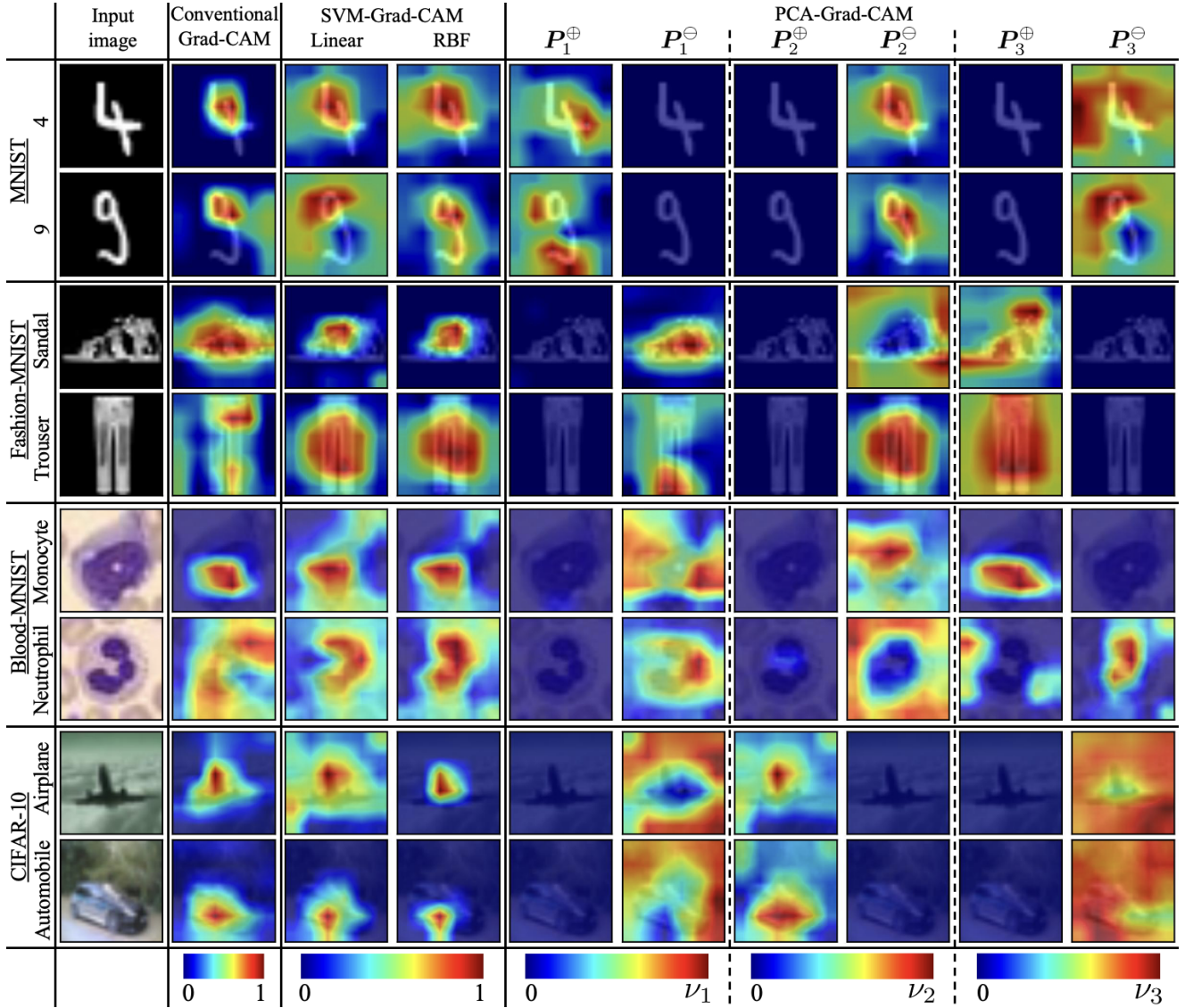


Fig. 8. Conventional-Grad-CAM, PCA-Grad-CAM, and SVM-Grad-CAM (MNIST [68], Fashion-MNIST [69], Blood-MNIST [70], and CIFAR-10 [66]).

the first principal component has a high contribution ratio, namely, the Chest X-Ray [71] [71] and Dog vs. Cat [72] models. As shown in Figure 6, since only the first principal component has a high contribution ratio, only P_1^+ and P_1^- were computed. Figure 9 indicates that PCA-Grad-CAM and SVM-Grad-CAM are highly similar. This likely indicates that the SVM performs classification using only the first principal component, p_1 . For the Chest X-Ray model, P_1^+ corresponds to “COVID-19” class, and P_1^- corresponds to “Pneumonia” class. It can be considered that regions associated with COVID-19 locally increase the first principal component, p_1 , whereas regions associated with pneumonia locally decrease p_1 . Similarly, for the Dog vs. Cat model, P_1^+ and P_1^- correspond to the faces of cats and dogs, respectively. It can be considered that the face of the cat locally increases the first principal component, p_1 , whereas the face of the dog locally decreases p_1 .

Previous studies showed that conventional Grad-CAM [8]

can visualize the focus regions of a CNN. However, when PCA or SVM is added to a CNN, the focus regions cannot be visualized. The proposed PCA-Grad-CAM and SVM-Grad-CAM solved this problem, so we believe this paper has certain value.

VII. CONCLUSION

In this study, we proposed a method to compute Grad-CAM for PCA and SVM layers embedded in a CNN. This allows us to identify the focus regions of PCA and SVM, which could not be observed with conventional methods (see Figures 8 and 9). In addition, using the gradient chain rule, we derived the closed-form expressions of the Jacobians required for PCA- and SVM-Grad-CAM (see Table II).

Note that there are many other dimensionality reduction and class estimation methods applicable to CNN features besides PCA and SVM. In the future, we aim to extend Grad-CAM to these methods as well. Although the proposed method is an

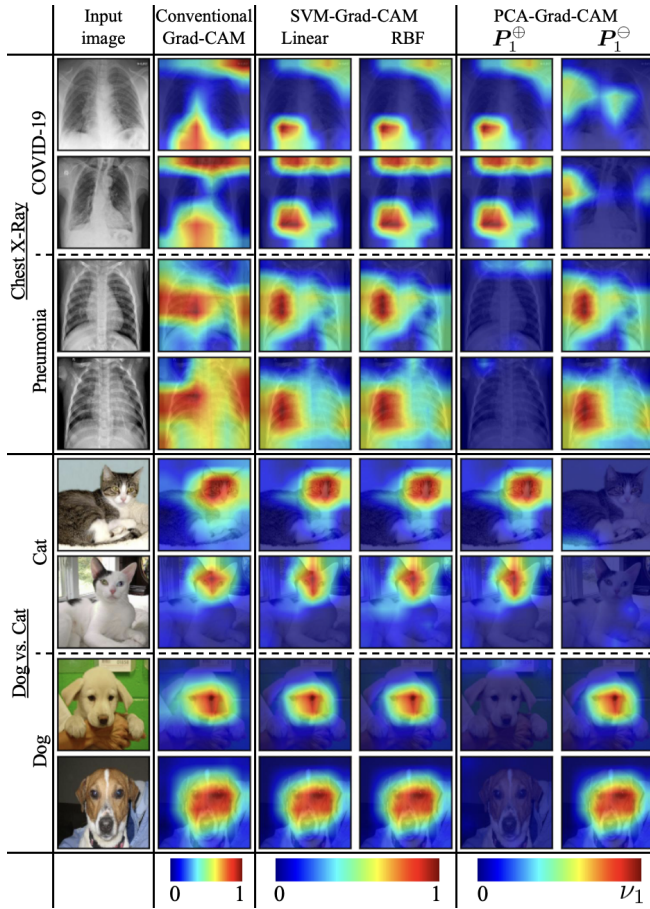


Fig. 9. Conventional-Grad-CAM, PCA-Grad-CAM, and SVM-Grad-CAM (Chest X-Ray [71] [71] and Dog vs. Cat [72]).

extension of Grad-CAM, the Jacobians derived in this study (see Table I) can also be directly used to visualize PCA and SVM layers in other first-order derivative-based visualization methods. Specifically, by slightly modifying the formulations of XGrad-CAM [30] and HiRes-CAM [38], these methods can also be applied to PCA and SVM layers. This will be reported in a future study. However, for Grad-CAM++ [26], which uses higher-order derivatives, and LIFT-CAM [58], which does not rely on derivatives, the PCA and SVM layers cannot be visualized using only the Jacobians. Extensions that allow the visualization of PCA and SVM layers for such methods are also needed. We plan to address this issue in future work.

APPENDIX A MATHEMATICAL REPRESENTATION

In this paper, a vector like $\zeta \in \mathbb{R}^Z$ is assumed to be a column vector. Then, the z -th component of ζ is denoted by $(\zeta)_z$, that is,

$$\zeta = [\zeta_1 \quad \dots \quad \zeta_Z]^\top \in \mathbb{R}^Z \Rightarrow (\zeta)_z = \zeta_z, \forall z \in \mathbb{N}_{\leq Z}.$$

Moreover, the gradient of the output vector $\eta = [\eta_1 \quad \dots \quad \eta_H]^\top \in \mathbb{R}^H$ with respect to the input vector $\zeta =$

$[\zeta_1 \quad \dots \quad \zeta_Z]^\top \in \mathbb{R}^Z$ is defined as

$$\frac{\partial \eta}{\partial \zeta} = \begin{bmatrix} \frac{\partial \eta_1}{\partial \zeta_1} & \dots & \frac{\partial \eta_1}{\partial \zeta_Z} \\ \vdots & \ddots & \vdots \\ \frac{\partial \eta_H}{\partial \zeta_1} & \dots & \frac{\partial \eta_H}{\partial \zeta_Z} \end{bmatrix} \in \mathbb{R}^{H \times Z}.$$

In this paper, we refer to it as the ‘‘Jacobian’’. We also define the scalar derivative of a vector and the vector derivative of a scalar as

$$\begin{aligned} \frac{\partial \eta}{\partial \zeta_z} &= \left[\frac{\partial \eta_1}{\partial \zeta_z} \quad \dots \quad \frac{\partial \eta_H}{\partial \zeta_z} \right]^\top \in \mathbb{R}^H, \\ \frac{\partial \eta_h}{\partial \zeta} &= \left[\frac{\partial \eta_h}{\partial \zeta_1} \quad \dots \quad \frac{\partial \eta_h}{\partial \zeta_Z} \right] \in \mathbb{R}^{1 \times Z}, \end{aligned}$$

respectively. By adopting this representation, the Jacobian can be expressed using

$$\frac{\partial \eta}{\partial \zeta} = \left[\frac{\partial \eta}{\partial \zeta_1} \quad \dots \quad \frac{\partial \eta}{\partial \zeta_Z} \right] = \begin{bmatrix} \frac{\partial \eta_1}{\partial \zeta} \\ \vdots \\ \frac{\partial \eta_H}{\partial \zeta} \end{bmatrix} \in \mathbb{R}^{H \times Z}.$$

APPENDIX B CLASS LABELS

- Fashion-MNIST [69] 0: T-shirt/top, 1: Trouser, 2: Pullover, 3: Dress, 4: Coat, 5: Sandal, 6: Shirt, 7: Sneaker, 8: Bag, 9: Ankle boot.
- Blood-MNIST [70] [75] 0: basophil, 1: eosinophil, 2: erythroblast, 3: immature granulocytes, 4: lymphocyte, 5: monocyte, 6: neutrophil, 7: platelet.
- CIFAR-10 [66] 0: airplane, 1: automobile, 2: bird, 3: cat, 4: deer, 5: dog, 6: frog, 7: horse, 8: ship, 9: truck.
- Chest X-ray [71] 0: COVID-19, 1: normal, 2: pneumonia.
- Dog vs. Cat [72] 0: cat, 1: dog.

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